

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397880
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Leslie; William D et al.

Accessory attachment system

Abstract

An accessory attachment system for quickly attaching and removing accessories on boats, vehicles and other objects in order to allow for temporary use of the accessory or accessories for a specific activity and/or application.

Inventors: Leslie; William D (Bradenton, FL), Kapper; Bradley J (Bradenton, FL)

Applicant: Leslie; William D (Bradenton, FL); Kapper; Bradley J (Bradenton, FL)

Family ID: 1000004825262

Assignee: EQUIPMOUNT LLC (Bradenton, FL)

Appl. No.: 16/856769

Filed: April 23, 2020

Publication Classification

Int. Cl.: B63B17/02 (20060101); B63B17/00 (20060101)

U.S. Cl.:

CPC B63B17/02 (20130101); B63B2017/0054 (20130101)

Field of Classification Search

CPC: B63B (17/00); B63B (17/02); B63B (2017/0054); B63H (21/00); B63H (21/10); A44B (1/00); A44B (1/34); A44B (17/00)

USPC: 114/343; 114/361; 114/364; 24/103; 24/104; 24/110

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4630982	12/1985	Fenner	24/701	B61D 45/001
4637492	12/1986	Herr	N/A	N/A
4966091	12/1989	De Suarez D'Aulan	N/A	N/A
5124902	12/1991	Puglisi	N/A	N/A
5268824	12/1992	Czipri	N/A	N/A
5803585	12/1997	Littman et al.	N/A	N/A
5927847	12/1998	Cales et al.	N/A	N/A
6089933	12/1999	Cotton	440/88M	B63B 13/02
6186088	12/2000	Garnier et al.	N/A	N/A
7021236	12/2005	Czipri	N/A	N/A
8122572	12/2011	Waddell	24/110	B63B 17/02
2008/0055905	12/2007	Chiu	N/A	N/A
2016/0288887	12/2015	Corley	N/A	N/A

Primary Examiner: Olson; Lars A

Attorney, Agent or Firm: LIVINGSTON LAW, P.A.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a continuation-in-part of application Ser. No. 16/004,820 filed on Jun. 11, 2018, which is currently pending. In addition, this application claims priority to application Ser. No. 62/808,017 filed on Feb. 20, 2019, which is currently pending. The patent applications identified above are incorporated herein by reference in their entirety to provide continuity of disclosure.

FIELD OF THE INVENTION

(2) This invention relates to boats and recreational vehicles, and more particularly a system for quickly attaching and removing accessories on boats, vehicles and other objects in order to allow for temporary use of the accessory for a specific activity and/or application.

BACKGROUND OF THE INVENTION

(3) There are many accessories available for boats and vehicles, such as automobiles, trucks, utility vehicles, all-terrain vehicles and so forth, which are used to customize a particular boat or vehicle for a specific activity. Such accessories are mounted onto a surface of the boat or vehicle. Attachable accessories are also available for other objects, such as trailers, work benches, tables and so forth.

(4) Attachable accessories for boats may include tie downs, towers, arches, bimini tops, fishing rod holders, seating, elevated decks, hunting blinds and so forth. Attachable accessories for vehicles may include storage racks, truck racks, utility racks, cargo carriers, tie downs, gun racks, extra seating, lights and so forth. Attachable accessories for other objects or work surface may include tools, vices, saws and so forth that are secured to a work surface prior to use.

(5) The conventional method of installing these types of accessories is to permanently attach the accessories to a desired surface. This causes a problem if an owner wants to use a boat, vehicle or

other object for multiple types of activities. For example, a boat owner may want to use a boat for fishing one weekend and then as a pleasure or cruising boat the next weekend, wherein certain accessories are not needed and will only take up space on the boat. A vehicle owner may want to use a storage rack for only the time necessary to transport cargo. Other individuals may want to temporarily secure a saw or other tool to a work surface only for the time necessary to perform a necessary task.

(6) Therefore, a need exists for an accessory attachment system for quickly attaching and removing accessories on boats, vehicles and other objects in order to allow for temporary use of the accessory for a specific activity and/or application.

SUMMARY OF THE INVENTION

(7) The primary object of the present invention is to provide an accessory attachment system for quickly attaching and removing accessories on boats, vehicles and other objects in order to allow for temporary use of the accessory for a specific activity and/or application.

(8) The present invention fulfills the above and other objects by providing an accessory attachment system for boats, vehicles and other objects wherein a socket is secured to a mounting surface and a corresponding stud is attached to an accessory, such as a tie down, anchor, shackle, shade supports, support poles, utility racks and so forth. The stud may then be removeably attached to the socket. A plurality of quick connects may be used for accessories having multiple attachment points. For example, the studs may support respective poles that support a canopy for shade.

(9) The accessory attachment system may also have a one or more mechanisms to allow for blind installations on surfaces wherein the underside of the surface is not accessible.

(10) The above and other objects, features and advantages of the present invention should become even more readily apparent to those skilled in the art upon a reading of the following detailed description in conjunction with the drawings wherein there is shown and described illustrative embodiments of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the following detailed description, reference will be made to the attached drawings in which:

(2) FIG. 1 is an exploded perspective side view of a quick connect of the accessory attachment system of the present invention;

(3) FIG. 2 is a sectional side view of a quick connect of the accessory attachment system of the present invention installed on a mounting surface;

(4) FIG. 3 is a perspective side view of a quick connect of the accessory attachment system of the present invention wherein the attachment bracket is a U-shaped attachment bracket;

(5) FIG. 4 is a side view of a boat having an accessory attached using the accessory attachment system of the present invention;

(6) FIG. 5 is an internal view of the boat along line A-A of FIG. 4 showing a quick connect of the system of the present invention;

(7) FIG. 6 is an exploded side view of a quick connect of the accessory attachment system of the present invention having a blind attachment means;

(8) FIG. 7 is an exploded perspective side view of a quick connect of the accessory attachment system of the present invention having a blind attachment means;

(9) FIG. 8 is a perspective side view of a quick connect of the present invention is illustrated wherein the stud and socket connect to each other via channels and pins;

(10) FIG. 9 is an exploded perspective side view of a quick connect of the present invention is illustrated wherein the stud and socket connect to each other via channels and pins;

(11) FIG. 10 is a top view of a stud of the present invention wherein the mounting surface has a

geometric shape;

(12) FIG. 11 is a side view of a stud of the present invention wherein the mounting surface has a geometric shape;

(13) FIG. 12 is a sectional side view along line B-B of FIG. 11 of a stud of the present invention wherein the mounting surface has a geometric shape;

(14) FIG. 13 is a side perspective view of a stud of the present invention wherein the mounting surface has a geometric shape;

(15) FIG. 14 is a perspective top view of a compression ring of the present invention;

(16) FIG. 15 is a perspective bottom view of a compression ring of the present invention;

(17) FIG. 16 is a bottom view of a compression ring of the present invention;

(18) FIG. 17 is a side view of a compression ring of the present invention;

(19) FIG. 18 is a sectional side view along line A-A of FIG. 17 of a compression ring of the present invention;

(20) FIG. 19 is a perspective top view of a socket of the present invention;

(21) FIG. 20 is a side plan view of the socket of the present invention;

(22) FIG. 21 is an alternate side plan view of the socket of the present invention;

(23) FIG. 22 is a sectional side view along line B-B of FIG. 21 of the socket of the present invention;

(24) FIG. 23 is a perspective top view of a bolt-on socket of the present invention;

(25) FIG. 24 is an exploded perspective top view of a bolt-on socket of the present invention;

(26) FIG. 25 is a perspective top view of a threaded socket of the present invention;

(27) FIG. 26 is an exploded perspective top view of a threaded socket of the present invention;

(28) FIG. 27 is a perspective top view of a socket of the present invention having a keyed aperture;

(29) FIG. 28 is a top view of a socket of the present invention having a keyed aperture;

(30) FIG. 29 is a side view of a socket of the present invention having a keyed aperture;

(31) FIG. 30 is a bottom view of a socket of the present invention having a keyed aperture;

(32) FIG. 31 is a perspective side view of a socket of the present invention having a support post;

(33) FIG. 32 is a side view of a socket of the present invention having a support post;

(34) FIG. 33 is an alternate side view of a socket of the present invention having a support post;

(35) FIG. 34 is a perspective top view of a utility rack of the present invention;

(36) FIG. 35 is a side view of an upright form a utility rack of the present invention;

(37) FIG. 36 is a perspective side view of an upright form a utility rack of the present invention;

(38) FIG. 37 is a side view of a shackle extending upward from a stud of the present invention;

(39) FIG. 38 is an opposing side view of a shackle extending upward from a stud of the present invention;

(40) FIG. 39 is a side view of a tie down mount assembly of the present invention;

(41) FIG. 40 is a perspective exploded view of a tie down mount assembly of the present invention;

(42) FIG. 41 is a side view of a boat a plurality of canopy support poles attached to the boat and supporting a canopy; and

(43) FIG. 42 is a side plan view of a canopy support pole of the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(44) For purposes of describing the preferred embodiment, the terminology used in reference to the numbered accessories in the drawings is as follows: 1. quick connect 2. socket 3. inner surface of socket 4. outer surface of socket 5. bottom edge of socket 6. bottom opening of socket 7. top edge of socket 8. top opening of socket 9. flange 10. socket nut 11. mounting surface 12. aperture 13. stud 14. inner surface of stud 15. outer surface of stud 16. bottom edge of stud 17. bottom opening of stud 18. top end of stud 19. bracket 20. pad 21. top surface of pad 22. bottom surface of pad 23. attachment bolt 24. distal end of attachment bolt 25. proximal end of attachment bolt 26. grip 27. collar 28. U-shaped attachment bracket 29. boat 30. accessory 31. arch 32. leg 33. end of leg 34. channel 35. pin 36. opening of channel 37. compression ring 38. lip 39. tie down 40. loop 41. base

of tie down 42. bottom surface of base 43. bolt 44. mounting plate 45. washer 46. keyed aperture 47. support post 48. utility rack 49. bed rail cap 50. upright 51. bottom end of upright 52. crossbar 53. sleeve 54. shackle 55. pin 56. tie down mount 57. aperture 58. canopy support pole 59. canopy 60. top end of canopy support pole 61. bottom end of canopy support pole 62. cord 63. opening 64. side wall 65. clip 66. top end of canopy support pole 67. bottom end of canopy support pole 68. cleat

(45) With reference to FIGS. 1 and 2, an exploded perspective side view and a sectional side view, respectively, of a quick connect **1** of the accessory attachment system of the present invention are illustrated. The quick connect **1** of the present invention comprises a tubular socket **2** having an inner surface **3**, a threaded outer surface **4**, a bottom edge **5** surrounding a bottom opening **6**, a top edge **7** surrounding a top opening **8** and a flange **9** extending outward from the top edge **7**. A socket nut **10** engages the threaded outer surface **4** of the socket **2** to secure the socket **2** to a mounting surface **11**, such as a boat deck or surface of a vehicle, by placing the socket **2** through an aperture **12** in the mounting surface **11** and then sandwiching the mounting surface **11** between the socket nut **10** and the flange **9**.

(46) Accessories are secured to at least one stud **13** that is inserted into the socket **2** in order to secure an accessory to a boat or vehicle to which the at least one socket **2** is attached. The stud **13** comprises a threaded inner surface **14**, an outer surface **15**, a bottom edge **16** surrounding a bottom opening **17**, a top end **18** and an attachment bracket **19** extending from the top end **18**. As illustrated in FIGS. 1 and 2, the attachment bracket **19** is a circular-shaped attachment pad **20** having a top surface **21** to which an accessory may be attached via welding or other attachment means. A bottom surface **22** of the pad **20** is preferably partially recessed to allow the flange **9** of the socket **2** to be fully covered by the pad **20**, as illustrated in FIG. 2.

(47) The stud **13** and socket **2** may have any corresponding geometric shape. As illustrated herein, each have a circular shape, however, each may have other geometric shapes, such as a rectangular, triangular, star, hexagonal, octagonal and so forth.

(48) The stud **13** is secured within the socket **2** via a threaded attachment bolt **23** having a distal end **24** that is inserted through the bottom opening **6** of the socket **2** and engages the threaded inner surface **14** of the stud **13**. The threaded attachment bolt **23** preferably has a grip **26** located on a proximal end **25** used for turning the threaded attachment bolt **23** and a collar **27** located below the grip **26** that engages the bottom edge **5** of the socket **2** to further secure the stud **13** within the socket **2**.

(49) With reference to FIG. 3, a perspective side view of a quick connect **2** of the accessory attachment system of the present invention wherein the attachment bracket **19** is a U-shaped attachment bracket **28** is illustrated. The stud **13** comprises a top end **18** and a bracket **19** extending from the top end **18** wherein the bracket is a U-shaped attachment bracket **28** to which an accessory may be attached using a pin or rivet.

(50) With reference to FIGS. 4 and 5, a side view of a boat **29** having an accessory **30** attached using the accessory attachment system of the present invention and an internal view of the boat **29** along line A-A of FIG. 4 showing a quick connect **2** of the system of the present invention, respectively, are illustrated. The accessory **30** is a support arch **31** secured to a deck mounting surface **11** of the boat **29** using a plurality of quick connects **2**. The arch **31** comprises legs **32** having ends **33** attached to attachment brackets **18**, which are pads **20**, of the quick connect **2** via welding or other attachment means. The legs **32** of the arch **31** are secured to the sockets **2** of the quick connects **1** by inserting the studs **13** into the corresponding sockets **2** and then securing the tubular shaped studs within the sockets **2** using threaded bolts.

(51) With reference to FIGS. 6 and 7, an exploded side view and exploded side perspective view, respectively, of a quick connect **1** of the accessory attachment system of the present invention having a blind attachment means **55** is illustrated. The quick connect **1** of the present invention comprises a tubular socket **2** having an inner surface **3**, a bottom edge **5** surrounding a bottom

opening **6**, a top edge **7** surrounding a top opening **8** and a flange **9** extending outward from the top edge **7**. At least one blind attachment means **55**, such as a button clip **56** or similar retractable projection **57**, is located on the perimeter wall of the tubular socket a predetermined distance from the flange **9**. To secure the socket **2** to a mounting surface **11**, such as a boat deck or surface of a vehicle, the socket **2** is placed through an aperture **12** in the mounting surface **11** while the at least one blind attachment means **55** is in a retracted position. When fully inserted into the aperture **12**, the blind attachment means **55** is released from the retracted position causing the mounting surface **11** to be sandwiched between the blind attachment means **55** and the flange **9**.

(52) As illustrated in FIG. **6**, the stud **13** may be secured within the socket **2** via at least one blind attachment means **55**, such as a button clip **56** or similar retractable projection **57** that engages an aperture located on the stud **13** or the bottom edge **5** of the socket **2**.

(53) As illustrated in FIG. **7**, the stud **13** may be secured within the socket **2** via at least one blind attachment means **55**, such as a button clip **56** or similar retractable projection **57** that engages an aperture or groove **58** located on the stud.

(54) With reference to FIGS. **8** and **9**, a quick connect **1** of the present invention is illustrated wherein the stud **13** and socket **2** connect to each other via channels **34** and pins **35** is illustrated. The quick connect **1** comprises a tubular-shaped socket **2** having an inner surface **3**, an outer surface **4**, a bottom edge **5** surrounding a bottom opening **6**, a top edge **7** surrounding a top opening **8** and a flange **9** extending outward from the top edge **7**. Pins **35**, as illustrated in FIGS. **20-22**, extend from the inner surface **3** of the socket **2** on opposing sides of the socket **2**. The quick connect **1** further comprises a stud **13** that is inserted into the socket **2** in order to secure an accessory to a surface to which the at least one socket **2** is attached. As further illustrated in FIGS. **10-13**, the stud **13** comprises an elongated cylindrical shaped body having, an outer surface **15**, a bottom edge **16**, a top end **18** and substantially J-shaped channels **34** located on the outer surface **15**. The J-shaped channels **34** have an opening **36** located on the bottom edge **16** of the stud **13**. The channels then extend upward toward the top end **18** of the stud **13** and curving or turning back toward the bottom edge **16** of the stud **13**. The stud **13** further comprises a compression ring **37**, as further illustrated in FIGS. **14-18**, located on the top end **18** of the cylindrical body of the stud **13** below a mounting surface **11** that extends outward from the outer surface **15** of the cylindrical body of the stud **13**, thereby creating a lip **38** that prevents the compression ring from sliding off the top end **18** of the stud **13**. The compression ring **37** may be constructed out of a rubber or equivalent material that provides resilience, such as a spring. As illustrated herein, the mounting surface **11** is hexagon-shaped to allow the stud **13** to be easily turned while pressing the stud **13** into the socket **2**.

(55) To couple the stud **13** and the socket **2**, the pins **35** on the stud **13** are aligned with the openings **36** of the channels **34** and the stud **13** is pushed into the socket **2**, thereby compressing the compression ring **37** between the flange **9** of the socket **2** and the lip **38** of the stud **13**. After the pins **35** reach the bottom of the channels **34**, the stud **13** is rotated so the pins slide along curved portions of the channels **34** and reach serifs **39** located at ends of the channels **34**. The compression ring **37** then pushes the stud **13** away from the socket **2** to keep the pins **35** locked in the serifs **39**.

(56) As further illustrated in FIGS. **8** and **9**, the accessory **30** attached to the stud **13** is a tie down **39** comprising a circular-shaped loop **40** extending from a base **41** having a concave hexagon-shaped bottom surface **42** that engages the hexagon-shaped mounting surface **11**. The mounting surface **11** and base **41** may have any corresponding geometric shapes. The tie down **39** is attached to the mounting surface **11** via a partially threaded bolt **43** that allows the tie down **39** to freely rotate in relation to the stud. Conversely, the tie down may be pushed downward to engage the mounting surface **11** and stud **13** when rotating the stud **13** within the socket **2**.

(57) With reference to FIGS. **19-22**, a socket **13** of the present invention is illustrated wherein the socket **2** comprises an inner surface **3**, an outer surface **4**, a bottom edge **5** surrounding a bottom opening **6**, a top edge **7** surrounding a top opening **8** and a flange **9** extending outward from the top

edge 7. Pins 35 extend from the inner surface 3 of the socket 2 on opposing sides of the socket 2. The socket 2 may be attached to a surface via an attachment means, such as welding.

(58) With reference to FIGS. 23 and 24, a bolt-on socket 13 of the present invention is illustrated wherein the socket 2 comprises an inner surface 3, an outer surface 4, a bottom edge 5, a top edge 7 surrounding a top opening 8 and a flange 9 extending outward from the top edge 7. Pins 35 extend from the inner surface 3 of the socket 2 on opposing sides of the socket 2. The socket 2 may be attached to a surface via bolts 43 that engage the flange 9 and nuts or a mounting plate 44.

(59) With reference to FIGS. 25 and 26, a threaded socket 13 of the present invention is illustrated wherein the socket 2 comprises an inner surface 3, an outer surface 4, a bottom edge 5, a top edge 7 surrounding a top opening 8 and a flange 9 extending outward from the top edge 7. Pins 35 extend from the inner surface 3 of the socket 2 on opposing sides of the socket 2. The outer surface 4 of the socket 2 is threaded so that it may be attached to a surface via a nut 10 and washer 45.

(60) With reference to FIGS. 27-30, a stud 13 having a keyed aperture 46 and comprising an elongated cylindrical shaped body having, an outer surface 15, a bottom edge 16, a top end 18 and substantially J-shaped channels 34 located on the outer surface 15. The J-shaped channels 34 have an opening 36 located on the bottom edge 16 of the stud 13. The channels then extend upward toward the top end 18 of the stud 13 and curving or turning back toward the bottom edge 16 of the stud 13. The keyed aperture 46 is located on the top end 18 of the stud 13 and allows accessories having similarly keyed male posts to be attached to the stud 13.

(61) With reference to FIGS. 31-33, a stud 13 having a support post 47 is illustrated. The stud 13 comprises an elongated cylindrical shaped body having, an outer surface 15, a bottom edge 16, a top end 18 and substantially J-shaped channels 34 located on the outer surface 15. The J-shaped channels 34 have an opening 36 located on the bottom edge 16 of the stud 13. The channels then extend upward toward the top end 18 of the stud 13 and curving or turning back toward the bottom edge 16 of the stud 13. The support post 47 is located on the top end 18 of the stud 13 and comprises a rectangular-shaped body having apertures 12 located thereon for locking an accessory to the support post 47.

(62) With reference to FIGS. 34-36, a utility rack 48 that incorporates quick connects 1 of the present invention is illustrated. The utility rack 48 comprises bed rail caps 49 that attach to bed rails of a pickup truck. Each bed rail cap 49 comprises an elongated member having a plurality of sockets 2 spaced apart along the length thereof. The sockets 2 provide attachment points for uprights 50. The uprights 50 comprises elongated members having studs 13 extending from the bottom ends 51 thereof. The studs 13 engage the sockets 2 of the bed rail caps 49 to lock the uprights 50 in vertical positions in relation to the bed rail caps 49. Crossbars 52 attach to the uprights 50 via sleeves 53 that allow the height of the crossbars 52 to be adjusted in relation to the uprights 50.

(63) With reference to FIGS. 37 and 38, a side view and opposing side view, respectively, of a shackle 54 extending upward from a stud 13 of the present invention are illustrated. The stud 13 comprises an elongated cylindrical shaped body having, an outer surface 15, a bottom edge 16, a top end 18 and substantially J-shaped channels 34 located on the outer surface 15. The J-shaped channels 34 have an opening 36 located on the bottom edge 16 of the stud 13. The channels then extend upward toward the top end 18 of the stud 13 and curving or turning back toward the bottom edge 16 of the stud 13. The stud 13 further comprises a compression ring 37 located on the top end 18 of the cylindrical body of the stud 13. The accessory 30 attached to the stud 13 is a shackle 54 comprising a substantially U-shaped loop 40 extending from a base 41 and connected to the base by a pin 55.

(64) With reference to FIGS. 39 and 40, a side view and perspective exploded view of a tie down mount 56 extending upward from a stud 13 of the present invention are illustrated. The stud 13 comprises an elongated cylindrical shaped body having, an outer surface 15, a bottom edge 16, a top end 18 and substantially J-shaped channels 34 located on the outer surface 15. The J-shaped

channels **34** have an opening **36** located on the bottom edge **16** of the stud **13**. The channels then extend upward toward the top end **18** of the stud **13** and curving or turning back toward the bottom edge **16** of the stud **13**. The stud **13** further comprises a compression ring **37** on the top end **18** of the cylindrical body of the stud **13** below a mounting surface **11** that extends outward from the outer surface **15** of the cylindrical body of the stud **13**, thereby creating a lip **38** that prevents the compression ring from sliding off the top end **18** of the stud **13**. As illustrated herein, the mounting surface **11** is hexagon-shaped to allow the stud **13** to be easily turned while pressing the stud **13** into the socket **2**.

(65) The accessory **30** attached to the stud **13** is a tie down mount **56** comprising a loop **40** having an aperture **57** located thereon for attaching a retractable tie down, such as a winch, ratcheting tie down and so forth. The loop **40** extends from a base **41** having a concave hexagon-shaped bottom surface **42** that engages the hexagon-shaped mounting surface **11**. The mounting surface **11** and base **41** may have any corresponding geometric shapes. The tie down mount **56** is attached to the mounting surface **11** via a partially threaded bolt **43** that allows the tie down mount **56** to freely rotate in relation to the stud. Conversely, the loop **40** may be pushed downward to engage the mounting surface **11** and stud **13** when rotating the stud **13** within the socket **2**.

(66) With reference to FIGS. **41** and **42**, a side view of a boat **29** having a plurality of canopy support poles **58** attached to the boat **29** and supporting a canopy **59** using the accessory attachment system of the present invention and a side plan view of a canopy support pole **58** of the present invention, respectively, are illustrated. Sockets **2** provide attachment points for canopy support poles **58** that extend upward to a desire height to support a canopy **59**, which provides shade over the boat **29** or other surface, structure or object having sockets **2** mounted thereon.

(67) The canopy support poles **58** are each preferably telescoping to allow the height of the canopy **59** to be adjusted. For example, one side of the canopy **59** may be made lower to provide additional shade from the sun in the early morning or in the late afternoon sun.

(68) Each canopy support pole **58** comprises a top end **60** and a bottom end **61**. A stud **13** is located on the bottom end **61** of the canopy support pole **58**. The stud **13** connects to a corresponding socket **2** via channels **34** and pins **35**, as previously illustrated. A cord **62** extends from said bottom end **61**, preferably entering through an opening **63** located in a side wall **64** of the canopy support pole **58**, through the canopy support pole **58** and preferably exiting through the top end **60** of the canopy support pole **58**. An attachment clip **65** is located on a top end **66** of said cord **62**. The attachment clip **65** may be secured to a corner or edge of the canopy **59** and then tightened by pulling a bottom end **67** of the cord **62**. The cord **62** may be tightened or loosened as the height of the canopy support pole **58** is adjusted, thereby adjusting the tautness of the canopy **59**. Finally, the cord **62** may be secured in place using a cleat **68** located on the canopy support pole **58**.

(69) It is to be understood that while a preferred embodiment of the invention is illustrated, it is not to be limited to the specific form or arrangement of parts herein described and shown. It will be apparent to those skilled in the art that various changes may be made without departing from the scope of the invention and the invention is not to be considered limited to what is shown and described in the specification and drawings.

Claims

1. An accessory attachment system comprising: a tubular-shaped socket having an inner surface, an outer surface, a bottom edge surrounding a bottom opening, a top edge surrounding a top opening and a flange extending outward from the top edge; at least one pin extending from the inner surface of the socket; a stud having an elongated cylindrical-shaped body capable of being inserted into said socket; said stud having an outer surface, a bottom edge, a top end and at least one channel located on the outer surface wherein said at least one channel engages said at least one pin when said stud is inserted into said socket; and an accessory extending from said top end of said stud

wherein said accessory is removeably attachable to said socket via said at least one stud wherein, said at least one channel located on said outer surface of said stud is a J-shaped channel having an opening located on the bottom edge of said stud; and said J-shaped channel extending upward toward the top end of said stud and turning back toward the bottom edge of the stud.

2. The accessory attachment system of claim 1 further comprising: a mounting surface located on the top end of said stud.

3. The accessory attachment system of claim 1 wherein: said accessory being a loop extending upward from the top end of said stud.

4. The accessory attachment system of claim 3 wherein: said loop having an aperture located thereon.

5. The accessory attachment system of claim 1 wherein: said accessory being a tie down extending upward from the top end of said stud.

6. The accessory attachment system of claim 1 wherein: said accessory being a shackle extending upward from the top end of said stud.

7. The accessory attachment system of claim 1 wherein: said accessory being a tie down mount extending upward from the top end of said stud.

8. An accessory attachment system comprising: a tubular-shaped socket having an inner surface, an outer surface, a bottom edge surrounding a bottom opening, a top edge surrounding a top opening and a flange extending outward from the top edge; at least one pin extending from the inner surface of the socket; a stud having an elongated cylindrical-shaped body capable of being inserted into said socket; said stud having an outer surface, a bottom edge, a top end and at least one channel located on the outer surface wherein said at least one channel engages said at least one pin when said stud is inserted into said socket; an accessory extending from said top end of said stud wherein said accessory is removeably attachable to said socket via said at least one stud; and a compression ring located on the top end of said stud.

9. The accessory attachment system of claim 8 further comprising: a mounting surface located on the top end of said stud above said compression ring; and said mounting surface extending outward from the outer surface of the cylindrical body of the stud thereby creating a lip that prevents the compression ring from sliding off the top end of the stud.

10. The accessory attachment system of claim 9 wherein: said mounting surface engages a base having a concave bottom surface with a geometric shape that corresponds to a shape of said mounting surface; and said base being located between said mounting surface and said accessory.

11. The accessory attachment system comprising: a tubular-shaped socket having an inner surface, an outer surface, a bottom edge surrounding a bottom opening, a top edge surrounding a top opening and a flange extending outward from the top edge; at least one pin extending from the inner surface of the socket; a stud having an elongated cylindrical-shaped body capable of being inserted into said socket; said stud having an outer surface, a bottom edge, a top end and at least one channel located on the outer surface wherein said at least one channel engages said at least one pin when said stud is inserted into said socket; and an accessory extending from said top end of said stud wherein said accessory is removeably attachable to said socket via said at least one stud, said accessory being an upright extending upward from the top end of said stud and supporting a crossbar.

12. The accessory attachment system comprising: a tubular-shaped socket having an inner surface, an outer surface, a bottom edge surrounding a bottom opening, a top edge surrounding a top opening and a flange extending outward from the top edge; at least one pin extending from the inner surface of the socket; a stud having an elongated cylindrical-shaped body capable of being inserted into said socket; said stud having an outer surface, a bottom edge, a top end and at least one channel located on the outer surface wherein said at least one channel engages said at least one pin when said stud is inserted into said socket; and an accessory extending from said top end of said stud wherein said accessory is removeably attachable to said socket via said at least one stud,

said accessory being a canopy support pole extending upward from the top end of said stud.

13. The accessory attachment system of claim 12 wherein: said canopy support pole having a top end and a bottom end; said canopy support pole being telescoping; a cord extending through the canopy support pole and exiting the top end of the canopy support pole; and an attachment clip for attaching the top end of the cord to a canopy.

14. The accessory attachment system of claim 13 wherein: a cleat located on the canopy support pole for securing said cord in a desired position.
